

be burned-in to a plasma screen. Both these display types are widely available in the market, and unless you are a videophile, picking up either of these technologies is a safe choice.

Rear-Projection TV

Microdisplays make up most rear-projection units. A microdisplay is driven by one or more microchips, which contain millions of pixels. Based on the nature of the chip, a microdisplay can be further classified as DLP, LCD, and LCoS. Each such unit displays by either bouncing light on to and off these chips (DLP and LCoS) or streaming light through them (LCD). The light source, being a lamp, must be replaced every 5,000 to 10,000 hours depending upon the technology and model (ask your local dealer for details); the lamps might also require a few seconds to warm up and cool down. Due to the nature of the technology, all such TVs are excellent for viewing in bright rooms. Let's quickly run through two of the most promising microdisplay technologies:

DLP: The microchip is actually an array of mirrors sitting on silicon, the number of mirrors correspond to the resolution of the set. These mirrors are rapidly repositioned to give various shades of grey by varying the intensity of light through a lens; black and white levels are achieved by switching the mirror off and on respectively. Colours are added by use of either a colour wheel, or by employing Red, Green, and Blue LEDs. A common complaint about DLP sets is that of the "rainbow effect"—this artefact is generally visible as a multi-colour shimmer when changing focus from one part of the projector screen to another, generally at a viewer's periphery when shifting focus from a high contrast area or bright object. The rainbow effect is mostly seen in single-chip DLP projectors. Make sure you check for this before buying a DLP set—people are not equally sensitive to the effect.

Some DLP sets use a technique called "wobulation" to double the actual resolution of a TV—a 1080p chip might have 960x1080 physical pixels. A mirror or lens is then made to move rapidly to form a complete picture, from the two halves of 960 pixels to give the 1920 x 1080 (1080p) finished picture. A wobulated image might appear soft; results vary from set to set.

LCoS: This is a technology similar to DLP, but uses liquid crystals instead of mirrors to reflect light. LCoS is the technology to watch—it has the potential to realise big-screen HDTVs with very high picture quality and at a low cost compared

Ideal Viewing Distance From An HDTV set

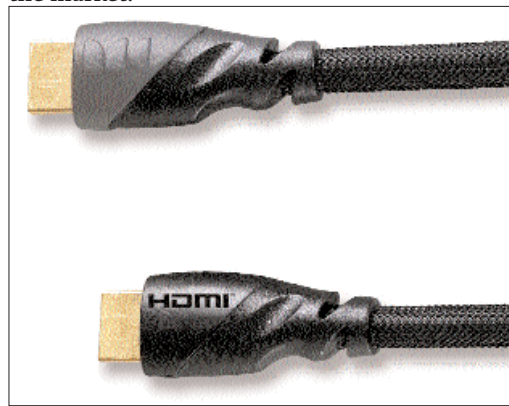
Screen size (inches)	Viewing distance (feet)
30	3.75 to 6.25
34	4.25 to 7
42	5.25 to 8.75
50	6.25 to 10.5
56	7 to 11.75
62	7.75 to 13
70	8.75 to 14.75



What Is An "HDTV-Ready" TV?

Some HDTVs might not have an inbuilt tuner to view over-the-air or broadcast HD content; these are generally labelled as "HDTV-ready TV" and must be connected to a separate HDTV set-top box to view them. Note that they will have the usual PAL/NTSC SDTV tuners.

to the others. Pixels on LCOS panels can be made smaller than is possible with other microdisplay technologies, without compromising on picture quality or manufacturability. LCoS displays can be scaled to 1080p resolution and beyond, without increasing the size and cost of the panel and other optical components in the light engine. Commercially, Sony's SXRD is the most visible and viable implementation of this technology in the market.



HDMI Cables

Summing It Up

For most of us, the fog of uncertainty surrounding HDTV lifts when we consider price as a factor. These big screens, sexy as they may be, are priced beyond EMIs for the rest of us. And when you look at HDTV through this lens of practicality, the choice becomes clear—whatever is the most affordable amongst this array of tech, and the most ubiquitous. Right now, that choice is a toss between Plasma and LCD sets. Too many companies have invested too much money for LCD to not have a slightly upper hand. LCD tech has overcome most of its inherent disabilities, and TV sets based on them are the cheapest money can buy today. Plasmas do not fall far outside an LCD's price bracket, and generally do make for a better home-theatre experience. But as far as the market is concerned, LCD is ascendant—you are bound to find a wide variety of prices and offers when shopping for an LCD-based HDTV.

Moreover, while some of the newer tech might or might not last tomorrow's dawn, LCD displays are here to stay. We'd recommend you wait a while before picking up an HD set—prices are bound to fall further, and 1080p displays should soon become more practical. By some major miracle, we might even see some HD content in India. But if you intend to walk that HD path this year, let LCD be your guiding light. ■

readersletters@jasubhai.com